# Phase 7 / Step 7.12: Molecular baseline (HD, NIST WebBook diatomic constants)

| | |
|---|---|
| State (ground) | X 1Σg+ 1sσ2 |
| T_e | 0 $cm^{-1}$ |
| ω_e | 3813.150000 $cm^{-1}$ (≈0.472770 eV) |
| ω_e x_e | 91.650000 $cm^{-1}$ (≈0.011363 eV) |
| B_e | 45.6550000 $cm^{-1}$ (≈0.00566050 eV) |
| α_e | 1.9860000 $cm^{-1}$ (≈0.00024623 eV) |
| D_e (distortion) | 0.0260500 $cm^{-1}$ (≈0.0000032298 eV) |
| r_e | 0.741420 Å (=7.414200e-11 m) |
| Morse (derived) | D_e≈4.9175 eV, D0≈4.6839 eV (from ωe, ωexe) |

Data: NIST Chemistry WebBook (Huber & Herzberg compilation)
This figure fixes baseline targets for Part III; it is not a P-model derivation.